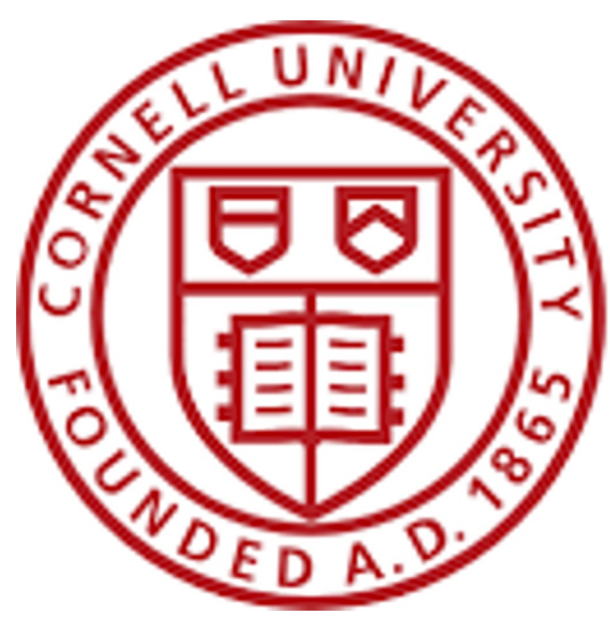




GPS said Nah. Wi-Fi said I got you

Anusha Muralidharan, Evan Navar Root, Erdis Brahim

Cornell University

MOTIVATION

Addressing GPS limitations

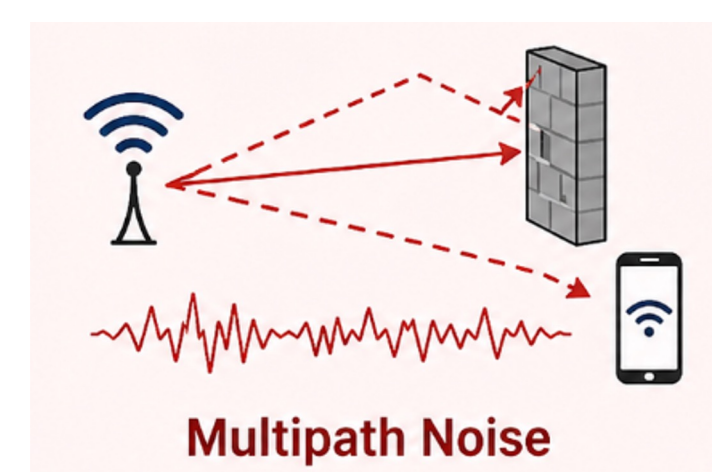


Wi-Fi is everywhere indoors

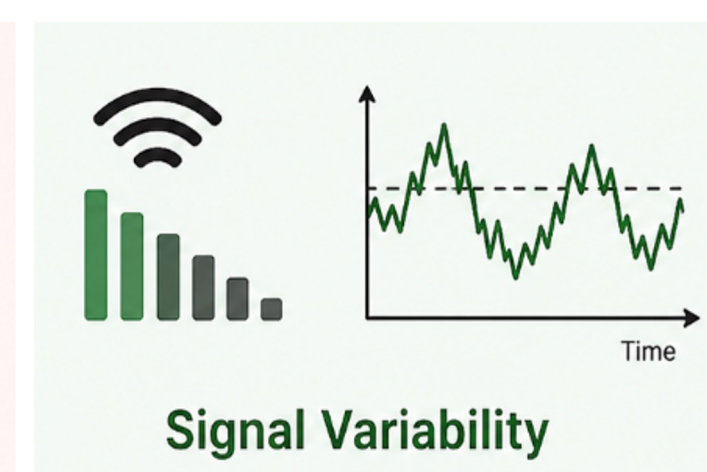


Infrastructure free deployment

CHALLENGES



Multipath Noise



Signal Variability



GPS Unavailable Indoors

GOAL

Learn location directly from radio frequency signals

DATASET

- Indoor room (1500 sq.ft) & 4 Access points(APs)
- ~10k samples per scenario

3 different scenarios

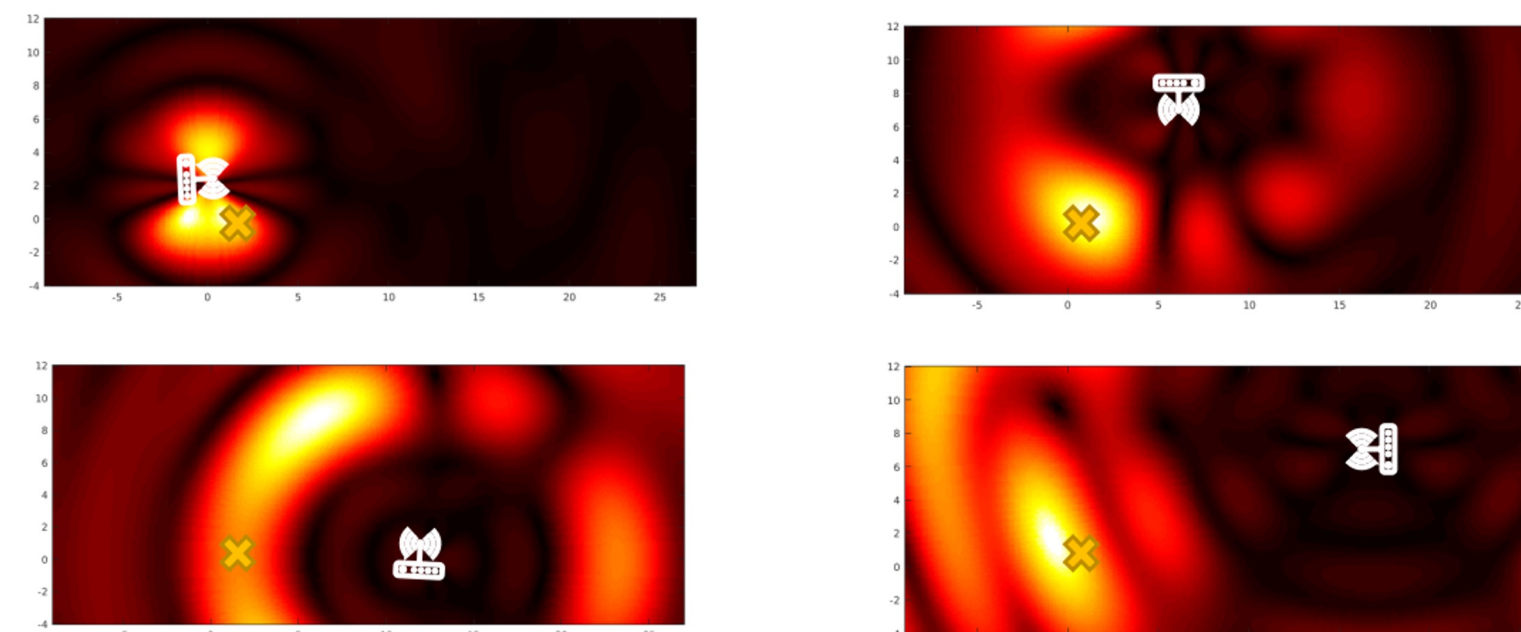


Ref: Ayyalasomayajula, Roshan, et al. "Deep learning based wireless localization for indoor navigation." Proceedings of the 26th Annual International Conference on Mobile Computing and Networking. 2020.

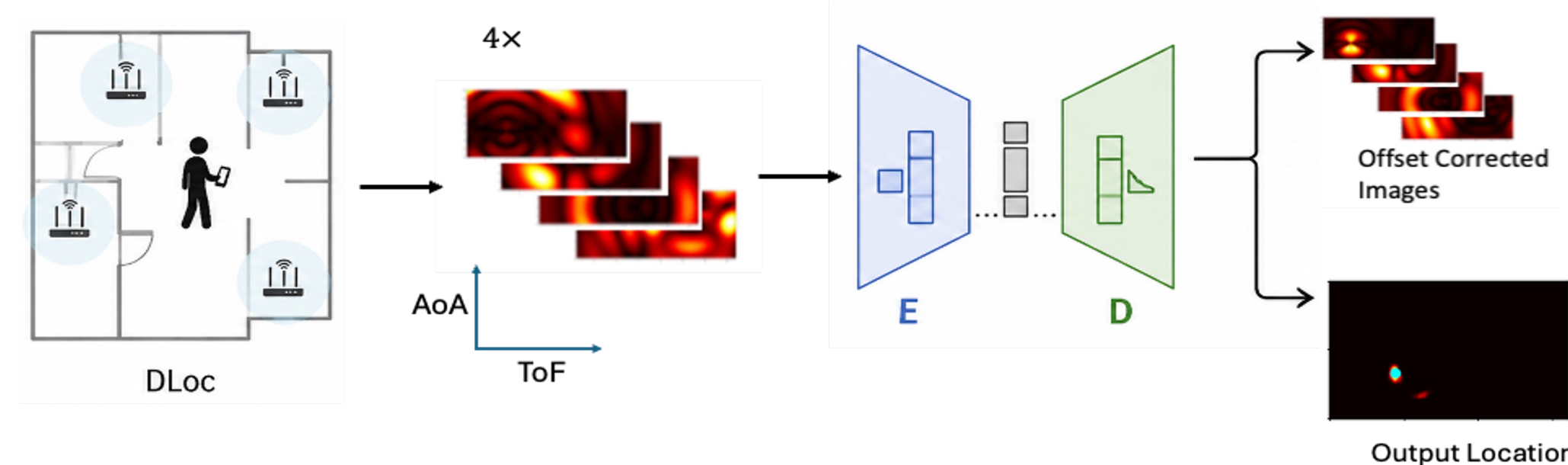
METHODOLOGY

INPUT

4 channel RF heatmaps encode spatial signature

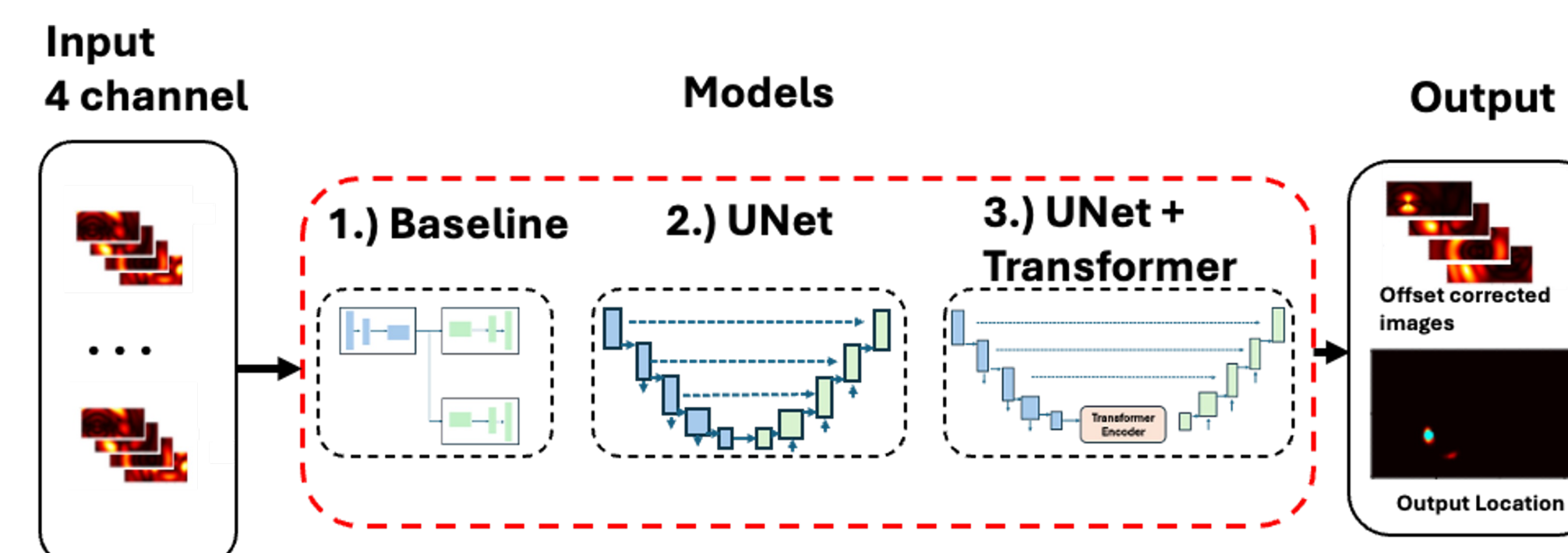


PIPELINE



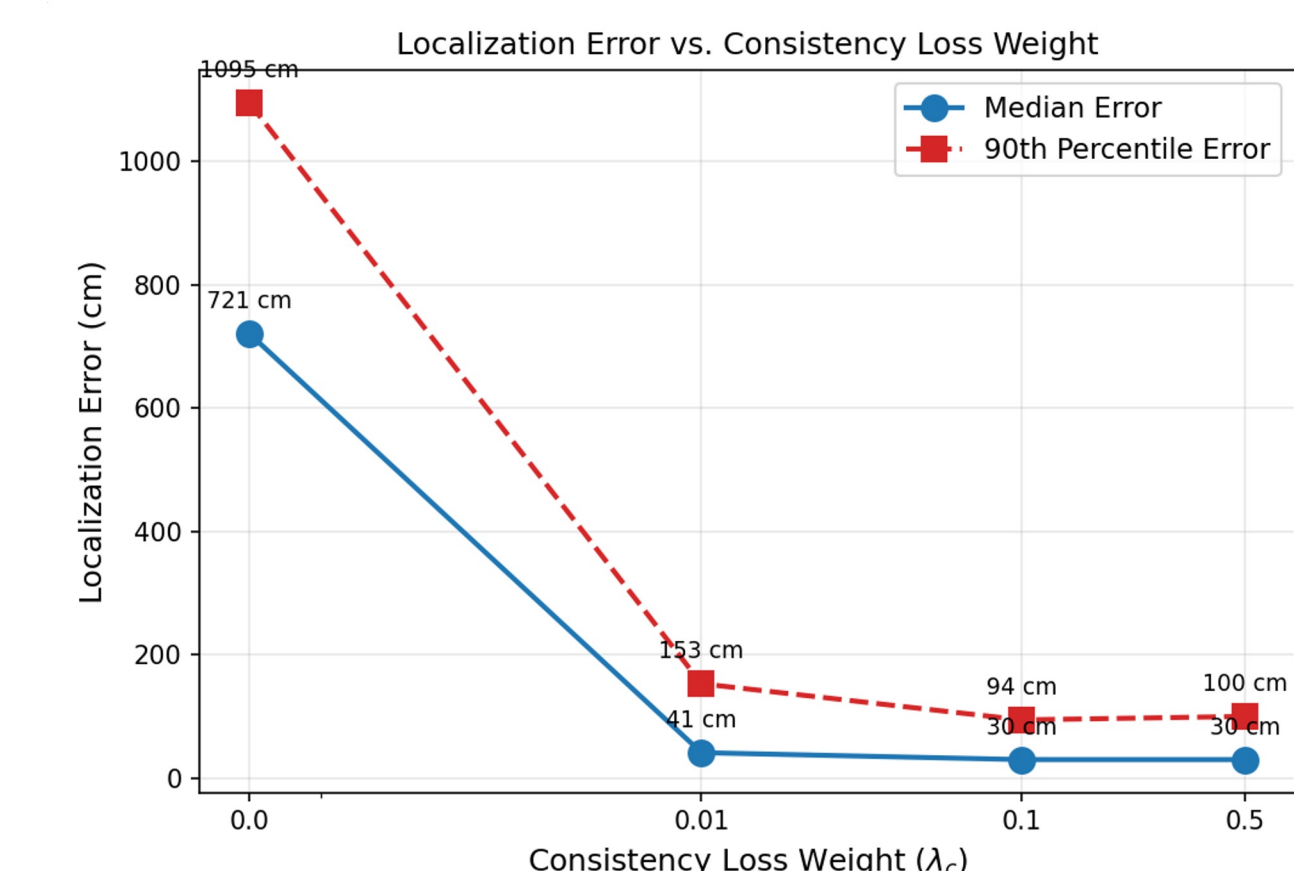
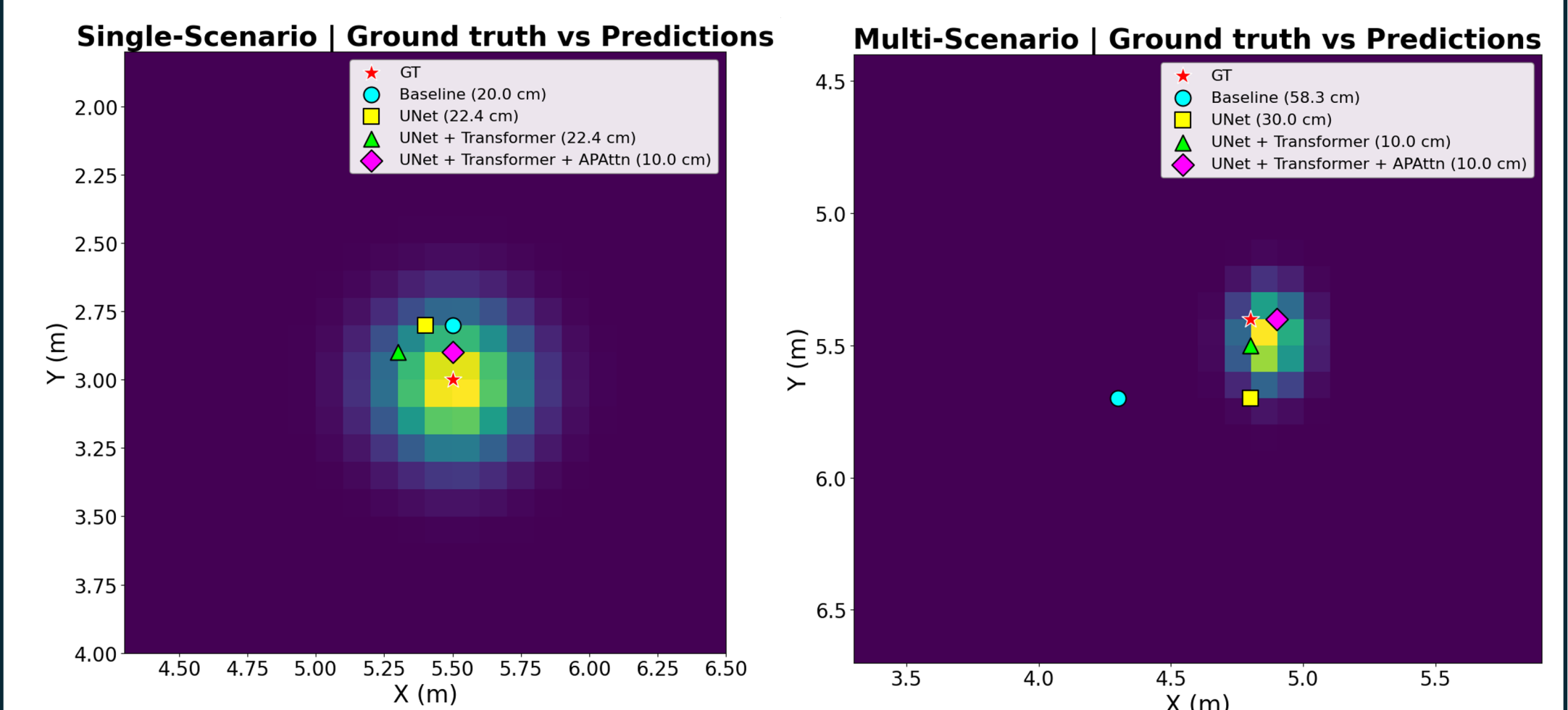
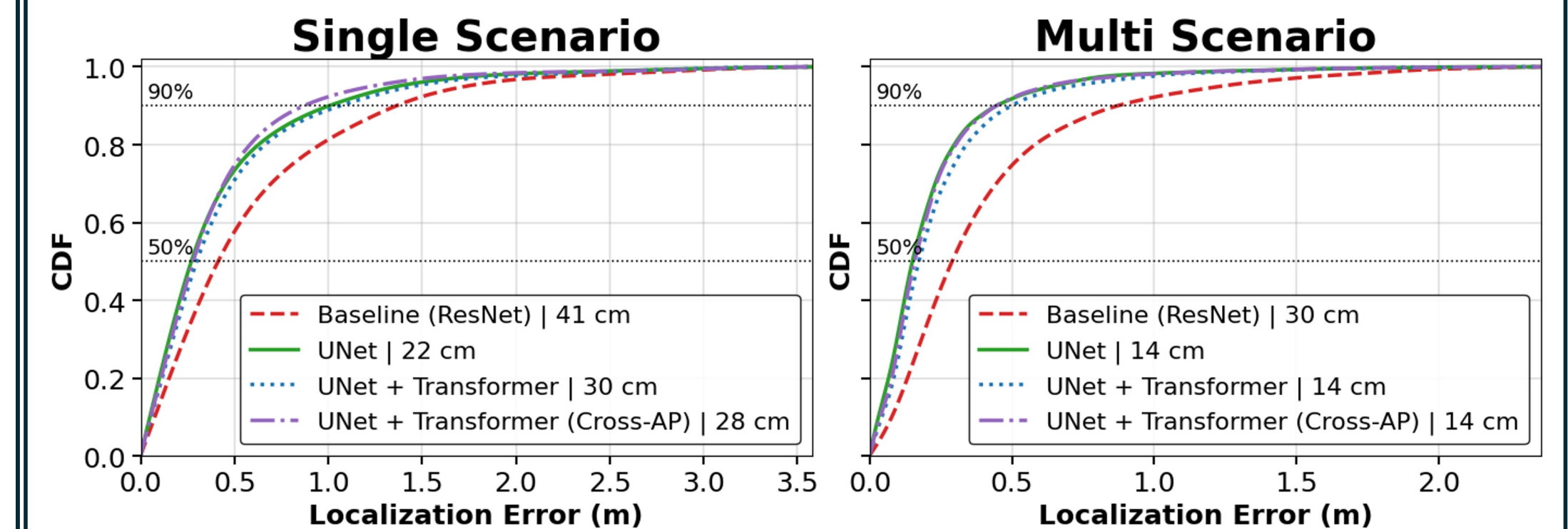
Consistency decoder corrects antenna effect
Location decoder predicts 2D location

MODELS



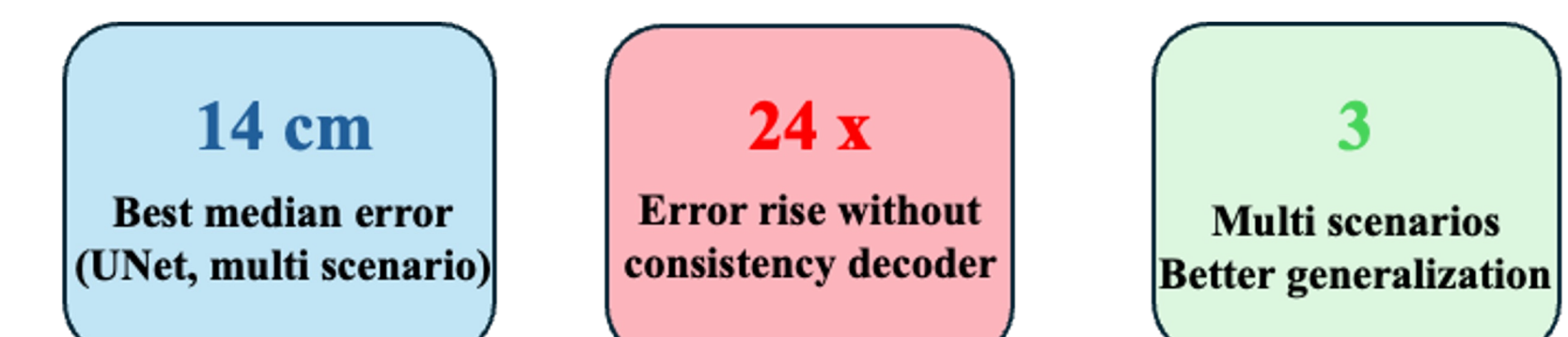
Model	Architecture feature	Parameters
Baseline	ResNet encoder decoder	18.8M
UNet	Skip connections	70.91M
UNet + Transformer	Attention at bottleneck	97.07M
UNet + Transformer+ AP Attn	Cross AP attention	97.09M

RESULTS



TAKEAWAYS

Reproduced original DLoc paper baseline results on the dataset



- Consistency decoder is critical for performance
- UNet achieves 80cm P90 (single) and 45cm P90 (multi)